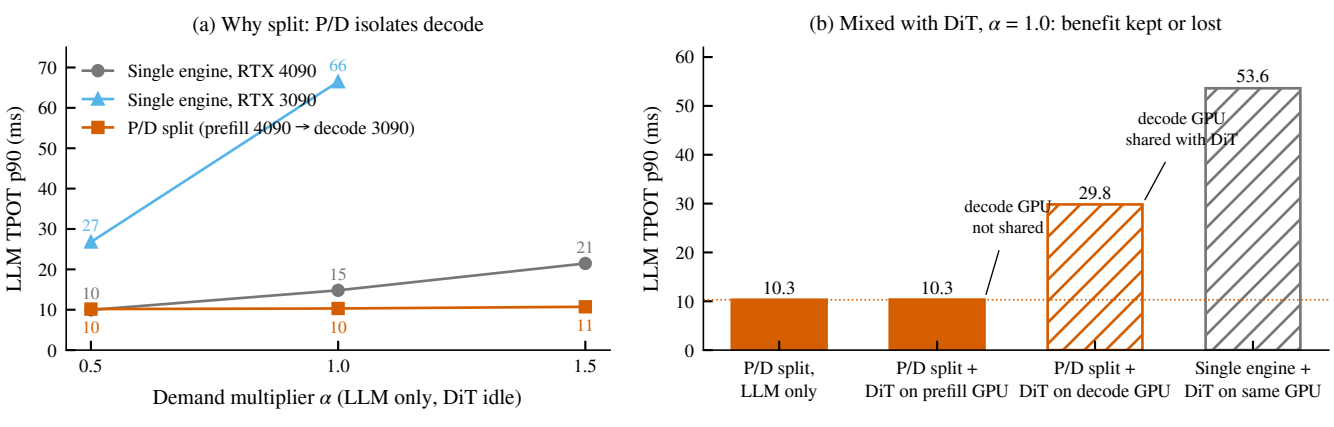
(a) Why split: P/D isolates decode
LLM TPOT p90 (ms)
Demand multiplier α (LLM only, DiT idle)
Single engine, RTX 4090
Single engine, RTX 3090
P/D split (prefill 4090 → decode 3090)
66
27
21
15
10
10
10
11
0
10
20
30
40
50
60
70
0.5
1.0
1.5
(b) Mixed with DiT, α = 1.0: benefit kept or lost
LLM TPOT p90 (ms)
10.3
10.3
29.8
53.6
decode GPU not shared
decode GPU shared with DiT
P/D split, LLM only
P/D split + DiT on prefill GPU
P/D split + DiT on decode GPU
Single engine + DiT on same GPU